

Smart Recruitment Matching Platform

1. Project Overview

Smart Recruitment Matching Platform is a secure web application that connects job seekers with employers. Job seekers create a profile and upload a CV, while employers post vacancies with the skills and experience they need. The system compares profile and CV skills against each job's requirements, produces a match score, and ranks candidates so employers see the most suitable applicants first. Job seekers can also see which required skills are missing from their profile, and the system prevents the same person from applying to the same vacancy more than once.

Basic User Flow

This is the only user flow supported by the system.

1. Register or log in as a job seeker or an employer
2. Job seeker completes a profile and uploads a CV; employer posts a job vacancy
3. System compares skills against the job requirements and creates a match score
4. Job seeker views matching jobs, checks any missing skills, and applies
5. Employer views the ranked applicant list and updates the application status
6. Employer sends a contact request to a candidate, and the candidate accepts or declines

2. Stakeholders

Stakeholder	Responsibility
Project Supervisor / Assessor	Guides the project and evaluates the final work.
Development Team	Designs, builds, and tests the application.
Job Seekers	Create profiles, upload CVs, and apply for jobs.
Employers	Post vacancies and review ranked applicants.
System Administrator	Manages user accounts and basic application settings.

3. User Roles

Role	Description
Job Seeker	Registers, logs in, manages a profile, uploads a CV, views matching jobs and missing skills, and applies for vacancies.
Employer	Registers, logs in, manages a company profile, posts and manages vacancies, views ranked applicants, updates application status, and sends contact requests.
Administrator	Manages user accounts and basic application settings and views a simple usage dashboard. The administrator does not change match results.

4. Proposed Technology Stack

Area	Proposed Technology
Front end	JavaScript, HTML, and CSS
Back end	ASP.NET Core Web API using C#
Database	SQL Server Express
Database tool	SQL Server Management Studio (SSMS)
Data access / ORM	Entity Framework Core preferred; ADO.NET as an alternative
Authentication	JWT-based authentication; ASP.NET Core Identity may be used if needed
Password security	Secure password hashing; never store passwords as plain text
Matching logic	Rule-based weighted scoring written in C# using skills, experience, education, and location
CV storage	Protected local server folder outside the public web folder

ASP.NET Core, SQL Server Express, and SQL Server Management Studio can all be used without paid licenses for a university project. SQL Server Express and SSMS are free tools provided by Microsoft.

5. Project Scope

In Scope

- User registration, login, and logout
- Secure user access using JWT
- Job seeker profile with skills, experience, education, and CV upload
- Employer company profile
- Posting, updating, and closing job vacancies
- Skill matching between job seeker profiles and job requirements
- Match score generation and candidate ranking
- Skill gap identification for job seekers
- Job search and basic filtering
- Applying for jobs and tracking application status
- Duplicate application prevention for the same vacancy
- Contact requests between employers and job seekers, with accept or decline
- In-application notifications for status changes
- Basic administrator account management and usage dashboard

Out of Scope

- Real-time chat or instant messaging
- Email and SMS notifications
- Online payments or paid job postings
- Machine learning or AI-based CV analysis
- Video interviews or online assessments
- Mobile application development
- Integration with external job boards

6. Functional Requirements

The following are the only functional requirements for this project.

No.	User Action	System Behaviour
1	User registers and logs in as a job seeker or an employer.	System validates the credentials and issues a JWT for secure access.
2	Job seeker creates and updates a profile with skills, experience, and education.	System saves the profile details and uses them for matching.
3	Job seeker uploads a CV.	System validates the file, stores it in the protected folder, and saves the metadata in the database.
4	Employer posts, updates, or closes a job vacancy.	System saves the vacancy with its required skills and experience.
5	Job seeker or employer opens a job or applicant list.	System compares skills against the job requirements and creates a match score.
6	Employer views applicants for a vacancy.	System displays the applicants ranked by match score, highest first.
7	Job seeker views a job.	System shows the match score and lists the required skills missing from the profile.
8	Job seeker searches and filters jobs.	System returns vacancies that match the search and filter values.
9	Job seeker applies for a vacancy.	System records the application and blocks a second application for the same vacancy.
10	Employer updates an application status.	System saves the new status and shows a notification to the job seeker.
11	Employer sends a contact request to a candidate.	System stores the request and allows the candidate to accept or decline it.
12	Administrator manages user accounts and opens the dashboard.	System allows account management and displays simple counts such as total users, vacancies, and applications.

7. Non-Functional Requirements

The following are the only non-functional requirements for this project.

Area	Requirement
Security	Passwords are hashed and all access is protected by JWT authentication. CV files are stored outside the public web folder.
Reliability	The system remains stable during normal use and handles errors without data loss.
Performance	Common actions such as login, job search, and match scoring respond within a few seconds.
Usability	The interface is simple and clear, so a new user can complete key tasks without training.
Matching accuracy	The same profile and vacancy always produce the same match score using the defined scoring rules.

8. Conclusion

This document defines the agreed business requirements for Smart Recruitment Matching Platform. The system will match job seeker skills against job requirements, rank candidates for employers, show job seekers their skill gaps, and prevent duplicate applications.